

Immediate vs delayed initiation of anti-TB therapy among patients hospitalised with HIV and suspected TB

Background

In patients hospitalised with HIV and suspected TB there is uncertainty around the benefits of early anti-TB therapy. The aim of this analysis was to estimate the effect of immediate vs. delayed initiation of anti-TB therapy.

Methods

Data

Data were analysed from a cohort of adults with a new suspected diagnosis of HIV-associated tuberculosis prospectively recruited between 2014 and 2016 from a hospital in Cape Town in South Africa. Further details of this cohort are provided in *Schutz et al. PLOS medicine. 2019*.¹

Inclusion criteria

In addition to the inclusion criteria that were applied for recruitment, additional inclusion criteria were required for this specific analysis. This analysis was restricted to individuals with microbiologically confirmed TB. Sputum, urine, and blood samples collected at enrolment were used to confirm TB status. Patients on anti-TB therapy at enrolment were excluded. Furthermore, patients who died on the enrolment day were excluded.

Treatment strategies

The treatment strategies under comparison were immediate initiation of anti-TB therapy on day 1 vs. delayed initiation within a grace period from 2 to 6 days following enrolment. Individuals were censored when non-adherent to the specified strategy (i.e., for the treatment strategy of delayed initiation at day 1 if anti-TB therapy was initiated on day 1, and at day 6 if therapy not initiated by that date).

Outcome

The outcome was all-cause death.

Covariates

Potential confounders were included and adjusted for in analyses. These were ascertained from measurements made and samples taken at enrolment. Covariates included as potential confounders were age, sex, days since hospitalisation, log creatinine, log CD4 count, log haemoglobin, impaired consciousness, hypoxia, rifampicin resistance, and pathogen load estimated using a graded response model.²

HIV viral load was included in descriptive statistics but was not adjusted for in analyses.

Follow-up

Follow-up was from enrolment until the earliest of death, loss to follow-up (LTFU), and 84 days following enrolment.

Causal contrasts

The causal contrast was the observational analogue of the per-protocol effect. Under the treatment strategies investigated, initiation of treatment was required but sustained use following initiation was not.

Statistical methods

Descriptive statistics were estimated for each treatment strategy. These included proportions for categorical covariates and medians with interquartile ranges for continuous covariates. Absolute standardised mean differences were used to quantify differences in covariates between treatment strategies.

Time was discretised in analyses into days. Risk of death at 84 days was estimated for each treatment strategy using weighted Kaplan-Meier estimators and compared using risk ratios and risk differences.³ Nonparametric bootstrapping with 1,000 resamples was used to calculate 95% confidence intervals by the percentile method.

Inverse probability weighting was used to account for censoring due to LTFU and for possible confounding due to common causes of delayed initiation and outcome. Censoring probabilities due to LTFU were estimated nonparametrically for each day among patients alive on that day using means estimated within levels of treatment initiated (yes/no) and for those who had initiated within levels of day of treatment initiation. Probabilities of immediate treatment initiation on day 1 were estimated using a logistic regression model adjusting for baseline covariates. Hazards of treatment initiation for days 2-6 were estimated among patients alive and not LTFU using a pooled logistic regression model adjusted for baseline covariates. Age was modelled in these logistic regression models using a restricted cubic spline. Days since hospitalisation was modelled using quadratic terms. Day was modelled in the pooled logistic regression model using a cubic polynomial. Weights were truncated at the 99th percentile.⁴

Secondary analyses

First, as a secondary analysis, risks were estimated within categories of estimated pathogen load. Pathogen load was categorised based on a graded response model into higher (>0) and lower (≤ 0) load.

Second, a naive analysis was conducted, vulnerable to immortal time bias, where the cohort is restricted to individuals who initiated treatment within 6 days. To initiate treatment after enrolment (e.g., at day 6) an individual has to survive until treatment receipt. Hence, requiring receipt of treatment can lead to bias in which estimates are biased towards reduced mortality with delayed treatment initiation.

Third, risks were estimated comparing a treatment strategy of immediate initiation vs. no immediate initiation (i.e., with no requirement for subsequent initiation within a grace period).

Sensitivity analyses

In sensitivity analyses, to assess the robustness of results to extreme weights, weights were not truncated or were truncated at the 90th or 95th percentile.

To assess the robustness of results to inclusion of individuals with antibiotic resistant TB, in a sensitivity analysis, individuals with rifampicin resistance were excluded.

Different weights can be used for grace period treatment strategies to imply different distributions of treatment during the grace period. The primary analysis used a natural grace period treatment strategy. As a sensitivity analysis, we estimated the effect of delayed initiation under a uniform distribution of treatment initiation within the grace period.^{5,6}

Preliminary descriptive statistics

Study population

From a total of 482 patients with confirmed TB, 64 exclusions were made due to the use of anti-TB therapy at baseline or due to death on the day of enrolment.

Table 1: Exclusions

Inclusion criteria	No. patients remaining	No. exclusions
None	482	NA
Not receiving anti-TB therapy at enrolment	419	63
No death on enrolment day	418	1

Among the remaining 418 patients, the number lost to follow-up was 7.

Time to anti-TB therapy initiation

Of the 418 included individuals, 397 initiated anti-TB therapy before death, loss to follow-up, or the end of follow-up (i.e., day 84). Most individuals initiated anti-TB therapy within the first six days following enrolment.

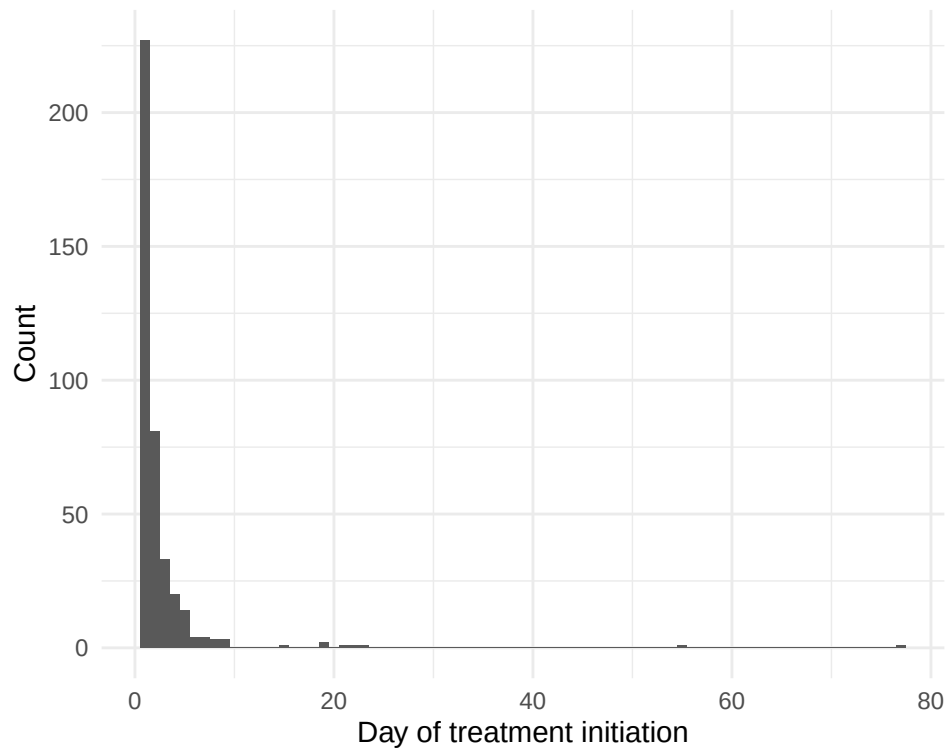


Figure 1: Histogram of day of treatment initiation

Baseline characteristics

Of 418 included individuals, 227 initiated anti-TB therapy on day 1 and 191 did not. Those who initiated immediately had on average lower haemoglobin levels and higher bacterial loads.

Table 2: Baseline characteristics by treatment initiation

Characteristic	Immediate initiation N = 227 ¹	Not immediate initiation N = 191 ¹	Difference ²
Female	113 (50%)	98 (51%)	-0.03
Age	36 (31, 44)	36 (31, 43)	-0.03
Days from hospital admission			0.06
0	10 (4.4%)	10 (5.2%)	
1	112 (49%)	90 (47%)	
2	40 (18%)	42 (22%)	
3	28 (12%)	25 (13%)	
4	21 (9.3%)	13 (6.8%)	
5	16 (7.0%)	11 (5.8%)	
HIV VL supressed	25 (12%)	21 (12%)	0.00
Unknown	11	12	
Log CD4 count	3.93 (2.89, 4.73)	3.95 (3.09, 4.72)	-0.05
Log haemoglobin	2.10 (1.92, 2.28)	2.19 (1.99, 2.34)	-0.24
Log creatinine	4.39 (4.16, 4.84)	4.41 (4.06, 4.84)	0.11
Rifampicin resistance	25 (11%)	22 (12%)	-0.02
Bacterial load GRM	0.27 (-0.46, 0.77)	-0.19 (-0.79, 0.66)	0.21
Impaired consciousness	33 (15%)	31 (16%)	-0.05
Hypoxia	17 (7.5%)	16 (8.4%)	-0.03

¹ n (%); Median (Q1, Q3)² Standardized Mean Difference

Weighted characteristics

To estimate the effect of the two treatment strategies, immediate vs. delayed initiation of anti-TB therapy, individuals were artificially censored when non-adherent to the assigned treatment strategy and weights were used to account for censoring.

After weighting, absolute standardised mean differences between those who followed a treatment strategy of immediate initiation relative to delayed initiation between 2 and 6 days following enrolment were small, and were all 0.1 or less.

Table 3: Weighted characteristics of individuals adhering to treatment strategies

Characteristic	Delayed initiation N = 378 ¹	Immediate initiation N = 418 ¹	Difference ²
Female	199 (53%)	213 (51%)	0.03
Age	36 (31, 42)	36 (31, 44)	-0.09
Days from hospital admission			-0.02
0	19 (4.9%)	18 (4.2%)	
1	171 (45%)	210 (50%)	
2	92 (24%)	75 (18%)	
3	53 (14%)	53 (13%)	
4	24 (6.5%)	35 (8.4%)	
5	19 (5.1%)	26 (6.3%)	
HIV VL supressed	39 (11%)	45 (11%)	-0.01
Unknown	21	21	
Log CD4 count	3.69 (2.94, 4.70)	3.99 (2.94, 4.79)	-0.05
Log haemoglobin	2.12 (1.92, 2.29)	2.12 (1.93, 2.29)	-0.10
Log creatinine	4.39 (4.06, 4.82)	4.37 (4.14, 4.72)	-0.07
Rifampicin resistance	38 (10.0%)	44 (11%)	-0.02
Bacterial load GRM	0.08 (-0.46, 0.82)	0.08 (-0.79, 0.77)	0.10
Impaired consciousness	61 (16%)	66 (16%)	0.01
Hypoxia	34 (9.0%)	32 (7.5%)	0.05

¹ n (%); Median (Q1, Q3)

² Standardized Mean Difference

The maximum of the truncated weights was less than 5.

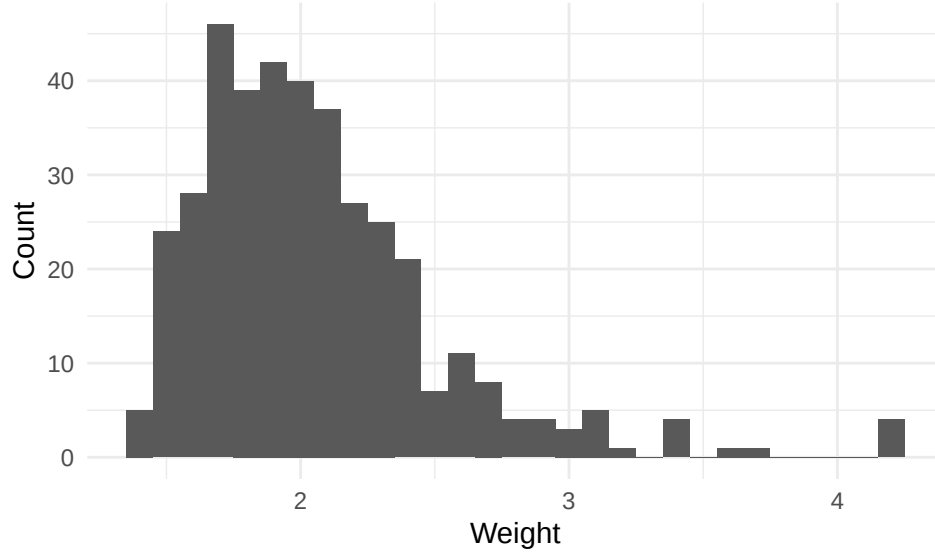


Figure 2: Histogram of weights at day 84

In bootstrap resampling, used to calculate confidence intervals, the maximum of the truncated weights for any bootstrap sample was less than 15.

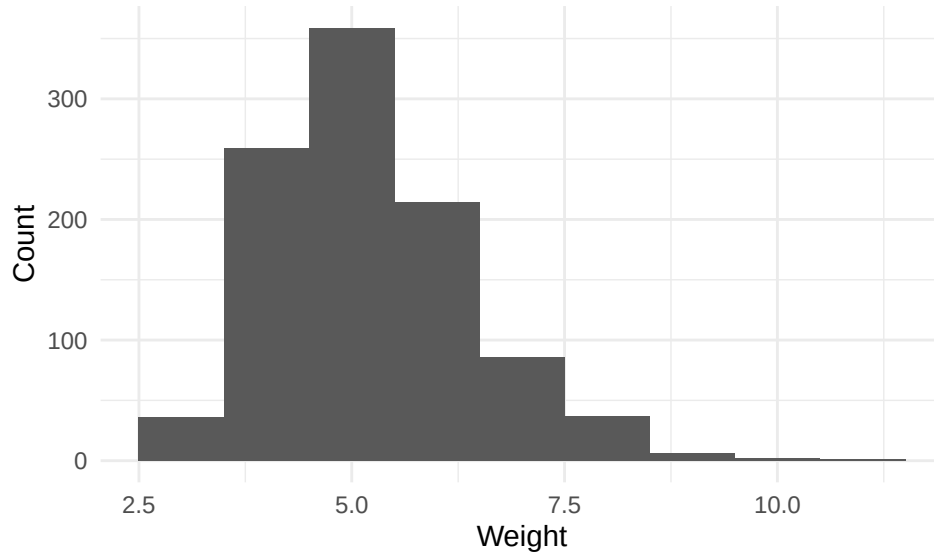


Figure 3: Histogram of the maximum weight at day 84 for each bootstrap sample

Primary analysis

Among 418 included individuals, there were 90 deaths.

Table 4: Crude death counts by treatment initiation

Treatment initiation	N	Deaths
Immediate initiation	227	42 (18.5%)
Not immediate initiation	191	48 (25.1%)

Risk of death with immediate initiation vs. delayed initiation within 2-6 days

The estimated risk of death by day 84 was 27.4% with delayed initiation and 17.9% with immediate initiation (RR 0.65, 95% CI 0.45, 0.96).

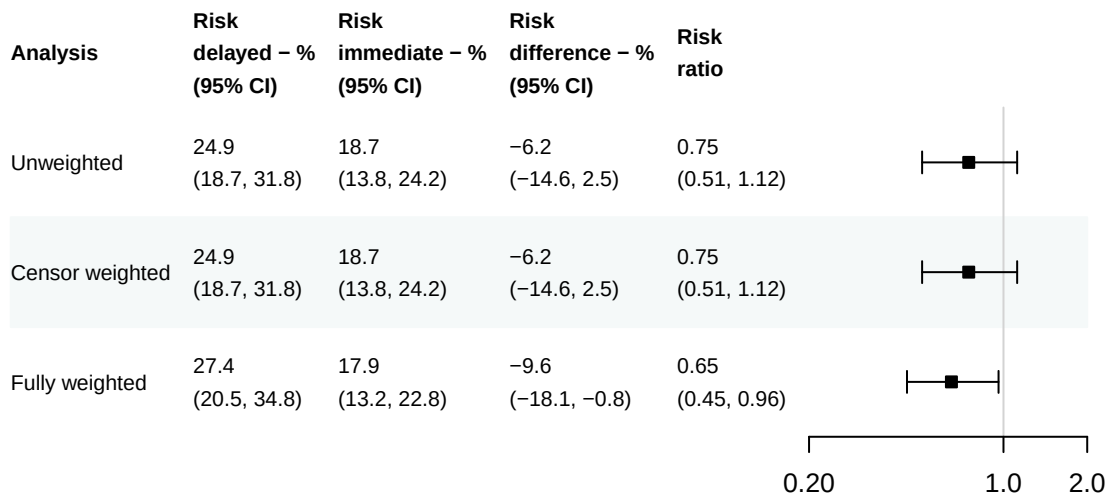


Figure 4: Weighted risks of death by day 84 with immediate versus delayed anti-TB therapy initiation

Note: Censor weighted analysis refers to an analysis with weighting only to account for censoring due to LTFU. The fully weighted analysis uses weighting for censoring due to LTFU and for confounding of timing of treatment initiation.

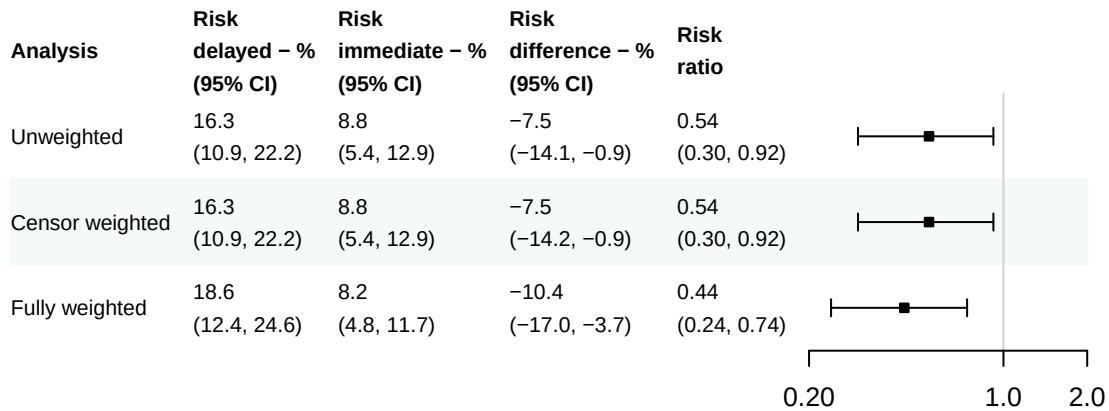


Figure 5: Weighted risks of death by day 28 with immediate versus delayed anti-TB therapy initiation

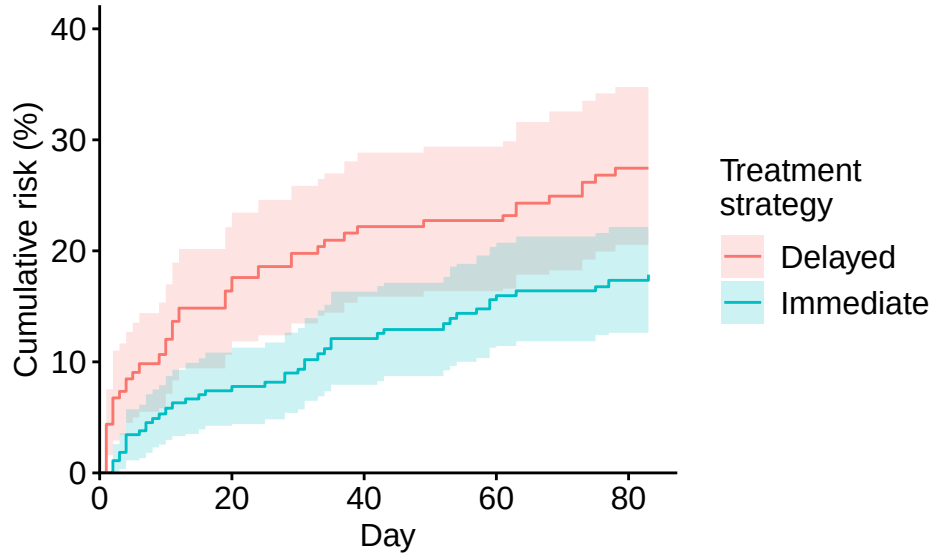


Figure 6: Weighted cumulative risk of death by treatment strategy

Secondary analysis

Differences in effect estimates by pathogen load

Of 418 patients, 207 (49.5%) had a bacterial load GRM score greater than 0 indicating higher pathogen load. Of those with a bacterial load GRM score above 0, 126 (60.9%) initiated immediately. Whereas of those with a bacterial load GRM score less than or equal to 0, 101 (47.9%) initiated immediately. The absolute risk difference was larger among individuals with a bacterial load GRM greater than 0 than those with values less than or equal to 0.

Table 5: Estimated risks of death by day 84 with immediate versus delayed anti-TB therapy initiation by GRM score

GRM score	Analysis	Risk with delayed initiation	Risk with immediate initiation	Risk difference	Risk ratio
Greater than 0	Unweighted	38.0 (28.0, 49.2)	22.3 (15.6, 29.7)	-15.8 (-29.1, -2.7)	0.59 (0.37, 0.91)
Greater than 0	Fully weighted	39.2 (28.5, 50.4)	21.1 (14.7, 28.5)	-18.1 (-30.6, -6.1)	0.54 (0.36, 0.81)
Less than or equal to 0	Unweighted	14.2 (7.8, 22.2)	14.2 (7.6, 21.5)	-0.1 (-9.7, 10.1)	1.00 (0.49, 2.10)
Less than or equal to 0	Fully weighted	13.9 (7.1, 21.4)	14.7 (8.2, 22.0)	0.8 (-9.0, 10.4)	1.06 (0.53, 2.16)

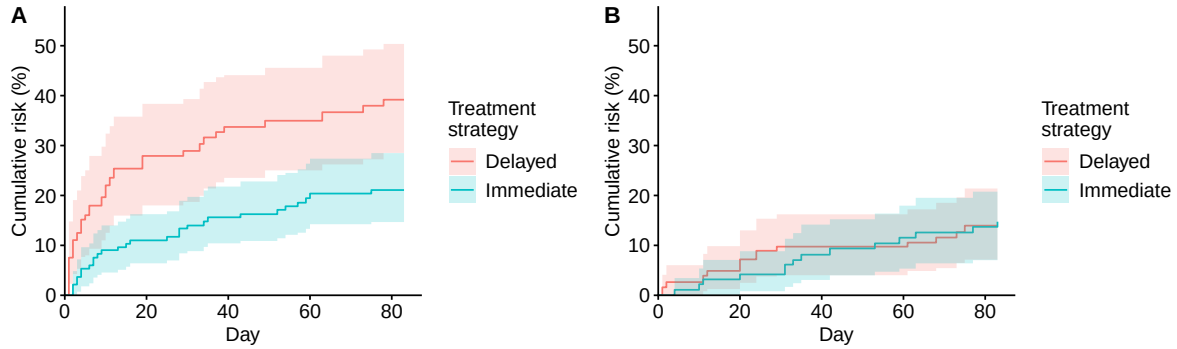


Figure 7: Weighted cumulative risk of death by treatment strategy for patients with A) a GRM score above 0 (higher pathogen load) and B) with a GRM score less than or equal to 0 (lower pathogen load)

Comparison to a naive approach

Rather than censor individuals when non-adherent to the specified regime, a naive approach would restrict to individuals who initiated therapy within days 1-6. This approach is susceptible to immortal time bias as those with delayed therapy initiation must necessarily have survived until the day of therapy initiation.

In this naive approach, relative risks were attenuated towards the null, as might be anticipated due to immortal time bias.

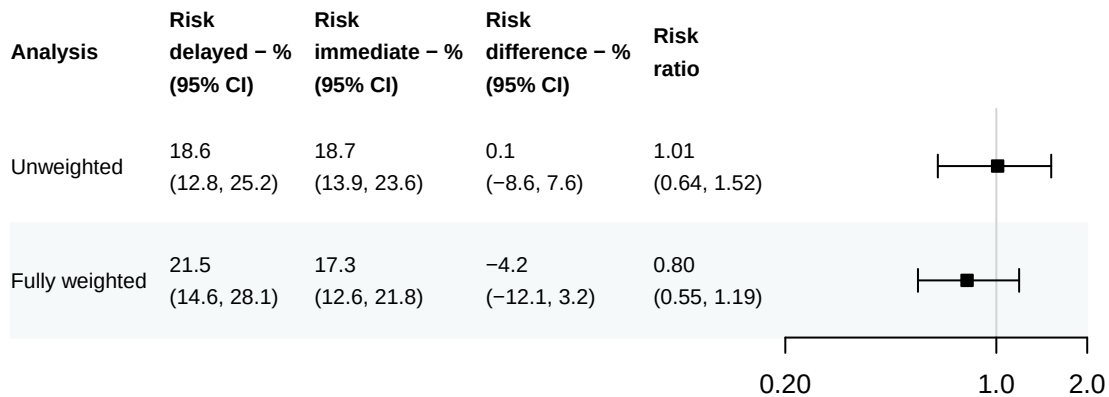


Figure 8: Estimated risks of death by day 84 with immediate versus delayed anti-TB therapy in naive approach

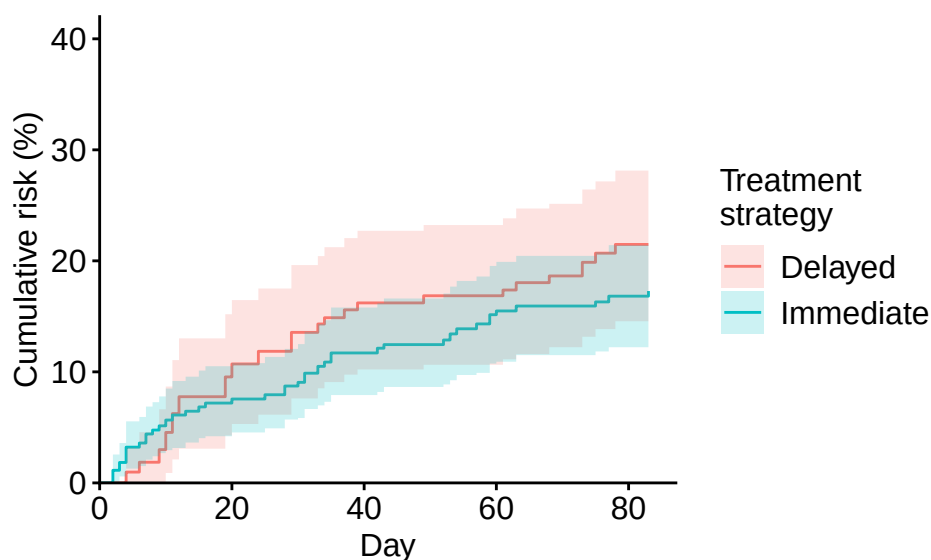


Figure 9: Weighted cumulative risk of death by treatment strategy using naive approach

Immediate initiation versus no immediate initiation

A simple comparator strategy to immediate initiation is no immediate initiation. No requirement is made in this treatment strategy of initiation by a given date. Results from this approach were similar to the primary analysis. The similarity is not unexpected given most individuals who do not initiate immediately do initiate within 6 days.

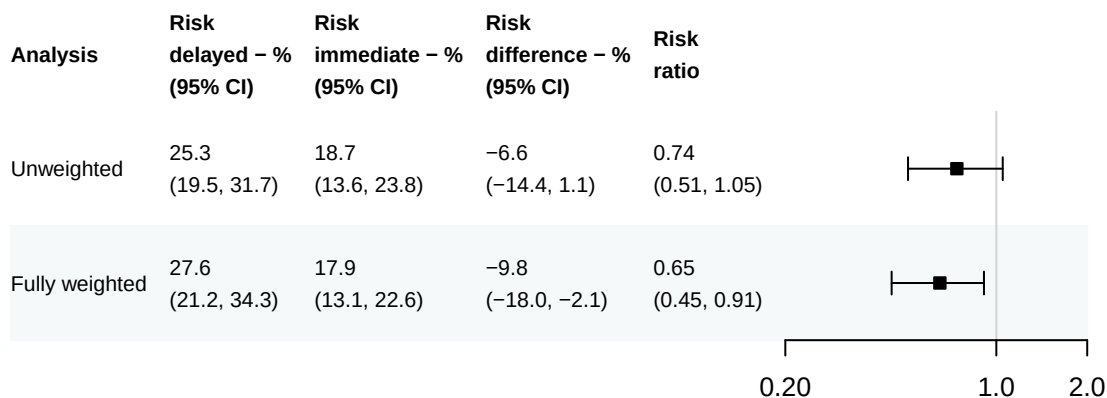


Figure 10: Estimated risks of death by day 84 with immediate versus no immediate anti-TB therapy

Sensitivity analyses

Weight truncation

Progressive truncation of the weights used in the primary analysis, from no truncation to truncation at the 90th percentile, did not result in substantive differences in effect estimates.

Table 6: Estimated risks of death by day 84 with immediate versus delayed anti-TB therapy with weight truncation

Analysis type	Analysis	Risk with delayed initiation	Risk with immediate initiation	Risk ratio	SE log(RR)
Immediate vs. 2-6 day delay - no truncation	Un-weighted	24.9 (18.5, 31.5)	18.7 (14.1, 23.9)	0.75 (0.50, 1.09)	0.200
Immediate vs. 2-6 day delay - no truncation	Fully weighted	27.7 (20.7, 35.5)	17.9 (13.3, 22.9)	0.64 (0.44, 0.94)	0.193
Immediate vs. 2-6 day delay - weights truncated at 99th percentile	Fully weighted	27.4 (20.9, 34.7)	17.9 (13.3, 22.9)	0.65 (0.44, 0.94)	0.189
Immediate vs. 2-6 day delay - weights truncated at 95th percentile	Fully weighted	27.1 (20.5, 33.9)	17.8 (13.4, 22.8)	0.66 (0.46, 0.95)	0.187
Immediate vs. 2-6 day delay - weights truncated at 90th percentile	Fully weighted	26.7 (20.4, 33.3)	17.8 (13.4, 22.7)	0.66 (0.46, 0.96)	0.188

Excluding those with rifampicin resistance

Rifampicin resistance may be both a confounder and an effect modifier for the effect of early vs. delayed therapy initiation. In a sensitivity analysis restricting to the 371 of 418 (88.8%) without rifampicin-resistant TB, relative risk estimates were compatible with the primary analysis though slightly closer to the null and with wider confidence intervals.

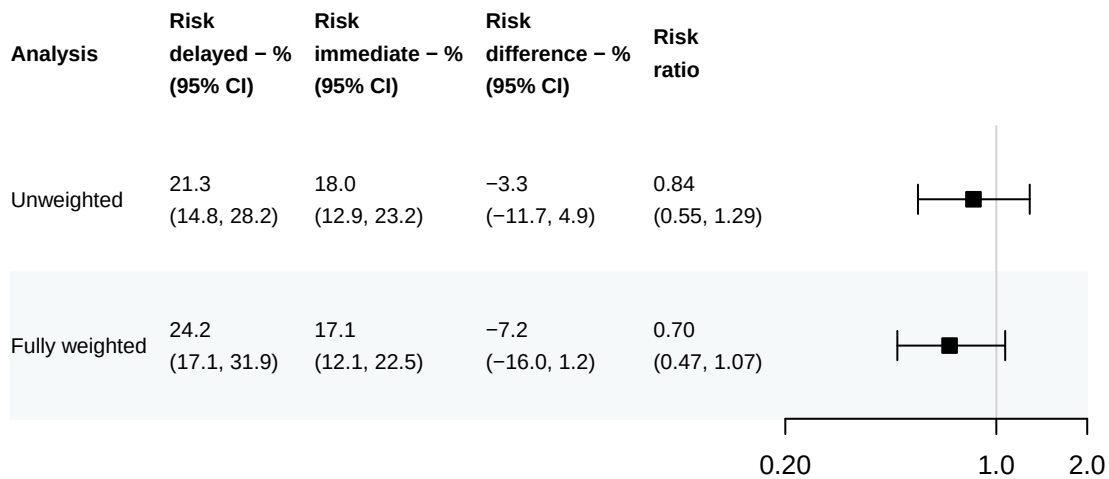


Figure 11: Estimated risks of death by day 84 with immediate versus delayed initiation of anti-TB therapy excluding individuals with rifampicin-resistant TB

Uniform grace period regime

Different weights can be applied during the grace period corresponding to different interventional distributions of treatment in the grace period. The treatment strategy of delayed initiation was a natural grace period regime as the interventional probability of therapy initiation within the interval was the natural (i.e., observed) probability of therapy initiation. As a sensitivity analysis, an alternative approach was used to specify a specific distribution of therapy initiation during the grace period (i.e., days 2 to 6), in which a uniform distribution of therapy initiation was specified. Estimated risks and risk ratios were similar to the primary analysis.

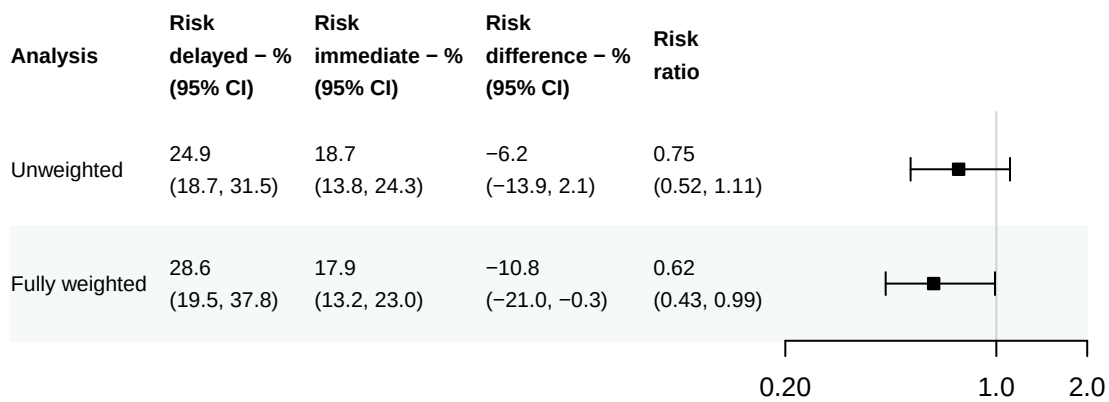


Figure 12: Estimated risks of death by day 84 with immediate versus delayed initiation of anti-TB therapy with a uniform grace period treatment strategy

Restricting to those enrolled within two days after day of hospitalisation

Patients were enrolled in the study within 0-5 days from the day of hospitalisation. As a sensitivity analysis, we restricted to those enrolled within two days after hospitalisation. Estimated risks and risk ratios were similar to the primary analysis but with wider confidence intervals.

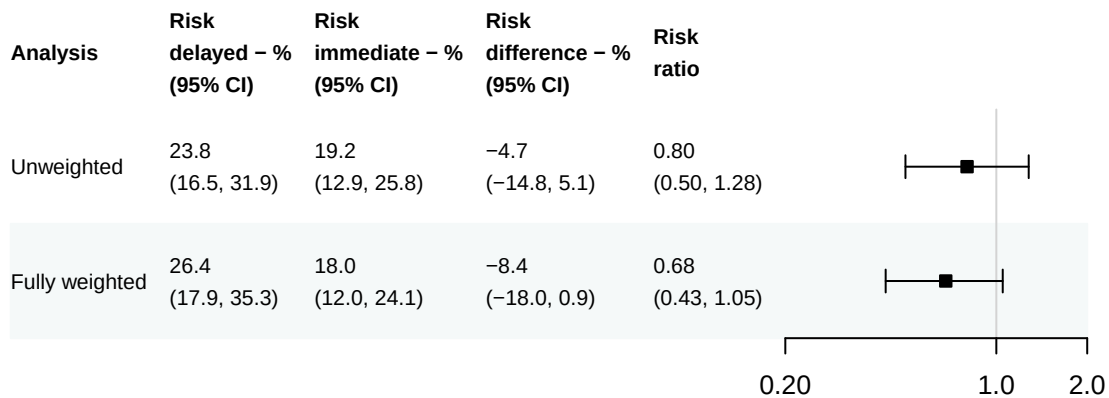


Figure 13: Estimated risks of death by day 84 with immediate versus delayed initiation restricted to those enrolled within 2 days after hospitalisation

References

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